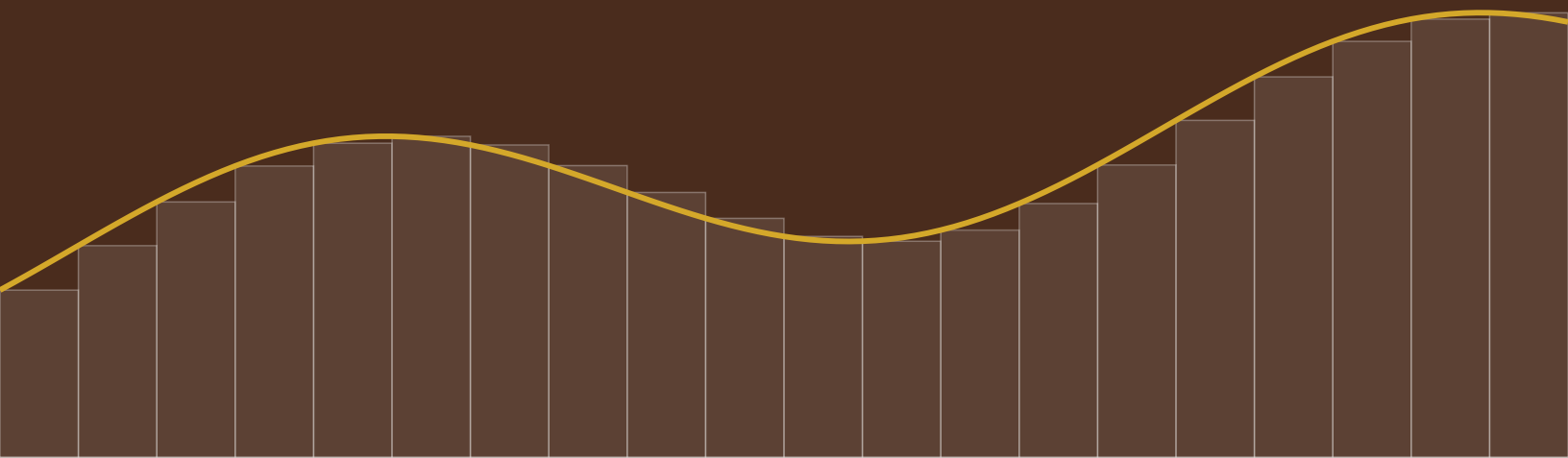




CARMEL CATHOLIC HIGH SCHOOL

AP Calculus BC

Course Notes

Based on Hass, Joel, et al. *Thomas' Calculus: Early Transcendentals*. 15th ed., Global ed., Pearson Education Limited, 2024.

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Functions

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NOTE This chapter is review from Precalculus. I think this chapter is mostly supplemental. But it's in the textbook and is important for the AP test.

1.1 Functions and Their Graphs

Functions are a tool for modeling the real world. They can be represented by an equation, a graph, a table, or a verbal description.

Functions; Domain and Range

In each case, the value of one variable depends on the value of another. In most cases, y is a function of x , symbolically represented as $y = f(x)$. The symbol f represents the function, the letter x is the input or, more accurately, *independent variable*, and y is the *dependent variable* or output value of f at x . A rule is classified as a function if and only if there is exactly one output for all possible inputs.

Definition 1.1.1 A function f from a set D to a set Y is a rule that assigns a single value $f(x)$ in Y to each x in D .

The set D represents all possible input values and is called the *domain*. The domain of f would sometimes be denoted by $D(f)$. The *natural domain* is the largest set of real x -values for which the formula gives real y -values.

The set of all output values $f(x)$ as x varies through D is called the *range* of the function. A function is said to be *real-valued* when its range is a set of real numbers.

Often a function is given by a formula that describes how to calculate the output value from the input variable. For instance, $y = x^2$.

Graphs of Functions

If f is a function with domain D , its *graph* contains the points in the Cartesian plane whose coordinates are the input-output pairs for f . In set notation the graph is $\{(x, f(x)) | x \in D\}$.

To graph a function, plot the points (x, y) that satisfy the equation on a Cartesian plane, and draw a smooth curve (labeled with its equation) through the plotted points.

Representing a Function Numerically

Another way functions can be represented is *numerically* through a table of values. Using this table, a graph of the function can be sketched. A graph containing only the points in the table is called a *scatterplot*.

The Vertical Line Test for a Function

A function can only have one value $f(x)$ for each x in $D(f)$. If a is in the domain of the function f , then the vertical line $x = a$ will intersect the graph of f at a single point $(a, f(a))$.

Piecewise-Defined Functions

Sometimes a function is described in pieces by using different formulas on different parts of its domain. It's written as

$$f(x) = \begin{cases} g(x) & x < -1 \\ h(x) - 1 & -1 \leq x < 1 \\ i(x) & x \geq 1 \end{cases}$$

Increasing and Decreasing Functions

Functions are classified based on how they respond to increasing inputs to understand the direction of change in a system.

Definition 1.1.2 Let f be a function defined on an interval I and let x_1 and x_2 be two distinct points in I .

1. If $f(x_2) > f(x_1)$ whenever $x_1 < x_2$, then f is said to be *increasing* on I .
2. If $f(x_2) < f(x_1)$ whenever $x_1 < x_2$, then f is said to be *decreasing* on I .

Even Functions and Odd Functions: Symmetry

Functions can be categorized by their geometric symmetry to simplify algebraic analysis and graphing.

Definition 1.1.3 Let f be a function and let $y = f(x)$

1. If $f(-x) = f(x)$ for all x in $D(f)$, then f is said to be an *even function* of x .
2. If $f(-x) = -f(x)$ for all x in $D(f)$, then f is said to be an *odd function* of x .

The graph of an even function is *symmetric about the y -axis*, and the graph of an odd function is *symmetric about the origin*. A graph is symmetric about the origin if a rotation of 180° about the origin leaves the graph unchanged.

Common Functions

A variety of important types of functions are frequently encountered in calculus.

Linear Functions

A *linear function* is a function of the form $f(x) = mx + b$, where m and b are constants. Its graph is a line with slope m and y -intercept b .

Linear functions are special because their rate of change is constant. The b 's cancel and the factor $x_2 - x_1$'s cancels, so the average rate of change between *any* two points is m . No other kind of function has this property.

FORMULA 1.1.1 (*Slope-intercept form*)

$$y = mx + b$$

Since the slope is constant, a single point (x_1, y_1) and the slope m can model the whole line. This is made simple with the point-slope form.

FORMULA 1.1.2 (*Point-slope form*)

$$y - y_1 = m(x - x_1)$$

Power Functions

A *power function* is a function of the form $f(x) = x^a$, where a is constant. Depending on the value of a , the function can behave in several ways.

In the case of $f(x) = x^a$, where a is a positive integer, the domain is all real numbers and the graph passes through $(0, 0)$ and $(1, 1)$. If a is even, the graph of the function is even, and its range is $[0, \infty)$. If a is odd, the graph of the function is odd, and its range is $(-\infty, \infty)$. As a increases, the graph gets flatter on $(-1, 1)$ and steeper for $|x| > 1$.

In the case of $f(x) = x^a$, where a is a negative integer, the function is $1/x^{-a}$ and 0 is excluded from its domain. If a is even, the graph of the function is even, increasing for negative x , decreasing for positive x , and its range is the positive real numbers. If a is odd, the graph of the function is odd and decreasing on each side of zero.

In the case of $f(x) = x^a$, where $a = p/q$ is a fraction in lowest terms, the function is $(\sqrt[q]{x})^p$. If q is even, the domain is the nonnegative real numbers, since even roots of negative numbers are

undefined. If q is odd, the domain is all real numbers. If p is even, the range is the nonnegative real numbers (positive if $a < 0$). If a is negative, zero is also excluded from the domain.

Polynomials

A *polynomial* is a function of the form $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, where n is a nonnegative integer, called the degree, and the numbers $a_0, a_1, \dots, a_n$ are real constants, called the coefficients. The domain is $(-\infty, \infty)$ for all polynomials.

Rational Functions

A *rational function* is a function of the form $f(x) = \frac{p(x)}{q(x)}$, where p and q are polynomials. The domain is all real numbers for which $q(x) \neq 0$.

Algebraic Functions

An *algebraic function* is a function constructed from polynomials using algebraic operations (addition, multiplication, division, taking roots). All rational functions are algebraic, but not all algebraic functions are rational.

Trigonometric Functions

The six basic *trigonometric functions* are functions that relate an angle to a ratio of side lengths in a right triangle. They are discussed further in Section 1.3.

Exponential Functions

An *exponential function* is a function of the form $f(x) = a^x$, where $a > 0$ and $a \neq 1$. They are discussed further in Section 1.4.

Logarithmic Functions

A *logarithmic function* is a function of the form $f(x) = \log_a x$, where the base $a \neq 1$ is a positive constant. They are the *inverse functions* of exponential functions. They are discussed further in Section 1.5.

Transcendental Functions

A *transcendental function* is any non-algebraic functions, such as the trigonometric functions, inverse trigonometric, exponential, logarithmic functions.

1.2 Combining Functions; Shifting and Scaling Graphs

Functions can be combined or transformed in many ways to form new functions and better model real-world situations.

Sums, Differences, Products, and Quotients

Functions can be added, subtracted, multiplied, and divided to produce new functions. Functions can also be multiplied by constants. Suppose f and g are functions and c is a real number.

Functions	Formula	Domain
$f + g$	$(f + g)(x) = f(x) + g(x)$	$D(f) \cap D(g)$
$f - g$	$(f - g)(x) = f(x) - g(x)$	$D(f) \cap D(g)$
$f \times g$	$(f \times g)(x) = f(x) \times g(x)$	$D(f) \cap D(g)$
f/g	$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$	$D(f) \cap D(g)$ at which $g(x) \neq 0$
$c \times f$	$(c \times f)(x) = c \times f(x)$	$D(f)$

The operator on the left-hand side represents an operation between functions while the operator on the right-hand side represents an operation between the real numbers $f(x)$ and $g(x)$.

Composing Functions

Functions can be combined through composition, where the output from one function becomes the input to another.

Definition 1.2.1 (*Function composition*) Suppose f and g are functions.

$$(f \circ g)(x) = f(g(x))$$

is the composition of f and g . The domain of $f \circ g$ consists of the numbers x in $D(g)$ for which $g(x)$ lies in the domain of f .

Shifting a Graph of a Function

Functions can be shifted by adding a constant to either the output of the existing function or the input variable.

FORMULA 1.2.1 (*Vertical Shifts*)

$$y = f(x) + k$$

Shifts the graph of f *up* k units if $k > 0$

Shifts the graph of f *down* $|k|$ units if $k < 0$

FORMULA 1.2.2 (*Horizontal Shifts*)

$$y = f(x + h)$$

Shifts the graph of f *left* h units if $h > 0$

Shifts the graph of f *right* $|h|$ units if $h < 0$

Scaling and Reflecting a Graph of a Function

Functions can be scaled by multiplying either the output of the existing function or the input variable by a constant c , where $c > 1$. Reflections across the coordinate axes are special cases where $c = -1$.

FORMULA 1.2.3 (*Vertical stretch*)

$$y = cf(x)$$

Stretches the graph of f vertically by a factor of c .

FORMULA 1.2.4 (*Vertical compression*)

$$y = \frac{1}{c}f(x)$$

Compresses the graph of f vertically by a factor of c .

FORMULA 1.2.5 (*Horizontal stretch*)

$$y = f(x/c)$$

Stretches the graph of f horizontally by a factor of c .

FORMULA 1.2.6 (*Horizontal compression*)

$$y = f(cx)$$

Compresses the graph of f horizontally by a factor of c .

FORMULA 1.2.7 (*Vertical reflection*)

$$y = -f(x)$$

Reflects the graph of f across the x -axis.

FORMULA 1.2.8 (*Horizontal reflection*)

$$y = f(-x)$$

Reflects the graph of f across the y -axis.

1.3 Trigonometric Functions

Angles

The Six Basic Trigonometric Functions

Periodicity and Graphs of the Trigonometric Functions

Trigonometric Identities

The Law of Cosines

Transformations of Trigonometric Graphs

Two Special Inequalities

1.4 Exponential Functions

Exponential Behavior

The Natural Exponential Function e^x

Exponential Growth and Decay

1.5 Inverse Functions and Logarithms

One-to-One Functions

Inverse Functions

Finding Inverses

Logarithmic Functions

Properties of Logarithms

Applications

Inverse Trigonometric Functions

The Arcsine and Arccosine Functions

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2.1 Rates of Change and Tangent Lines to Curves

Average and Instantaneous Speed

Defining the Slope of a Curve

Average Rates of Change and Secant Lines

Rates of Change and Tangent Lines

2.2 Limit of a Function and Limit Laws

An Informal Description of the Limit of a Function

Limits of Function Values

The Limit Laws

Eliminating Common Factors from Zero Denominators

Evaluating Limits of Polynomials and Rational Functions

Using Calculators and Computers to Estimate Limits

The Sandwich Theorem

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Approaching a Limit from One Side

Limits at Endpoints of an Interval

Precise Definitions of One-Sided Limits

Limits Involving $(\sin \theta)/\theta$

2.5 Limits Involving Infinity; Asymptotes of Graphs

Finite Limits as $x \rightarrow \pm\infty$

Limits at Infinity of Rational Functions

Horizontal Asymptotes

Oblique Asymptotes

Infinite Limits

Precise Definitions of Infinite Limits

Vertical Asymptotes

Dominant Terms

2.6 Continuity

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Continuous Functions

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3.1 Tangent Lines and the Derivative at a Point

Finding a Tangent Line to the Graph of a Function

Rates of Change: Derivative at a Point

3.2 The Derivative as a Function

Calculating Derivatives from the Definition

Notation

Graphing the Derivative

Differentiability on an Interval; One-Sided Derivatives

When Does a Function Not Have a Derivative at a Point?

Differentiable Functions Are Continuous

3.3 Differentiation Rules

Powers, Multiples, Sums, and Differences

Derivatives of Exponential Functions

Products and Quotients

Second- and Higher-Order Derivatives

3.4 The Derivative as a Rate of Change

Instantaneous Rates of Change

Motion Along a Line: Displacement, Velocity, Speed, Acceleration, and Jerk

Derivatives in Economics and Biology

Sensitivity to Change

3.5 Derivatives of Trigonometric Functions

Derivative of the Sine Function

Derivative of the Cosine Function

Simple Harmonic Motion

Derivatives of the Other Basic Trigonometric Functions

3.6 The Chain Rule

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“Outside-Inside” Rule

Repeated Use of the Chain Rule

The Chain Rule with Powers of a Function

3.7 Implicit Differentiation

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3.8 Derivatives of Inverse Functions and Logarithms

Derivatives of Inverses of Differentiable Functions

Derivative of the Natural Logarithm Function

The Derivatives of a^x and $\log_a x$

Logarithmic Differentiation

Irrational Exponents and the Power Rule (General Version)

The Number e Expressed as a Limit

3.9 Inverse Trigonometric Functions

Inverses of $\tan x$, $\cot x$, $\sec x$, and $\csc x$

The Derivative of $y = \arcsin u$

The Derivative of $y = \arctan u$

The Derivative of $y = \operatorname{arcsec} u$

Derivatives of the Other Three Inverse Trigonometric Functions

3.10 Related Rates

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3.11 Linearization and Differentials

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4.1 Extreme Values of Functions on Closed Intervals

Local (Relative) Extreme Values

Finding Extrema

4.2 The Mean Value Theorem

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The Mean Value Theorem

A Physical Interpretation

Mathematical Consequences

Finding Velocity and Position from Acceleration

4.3 Monotonic Functions and the First Derivative Test

Increasing Functions and Decreasing Functions

First Derivative Test for Local Extrema

4.4 Concavity and Curve Sketching

Concavity

Points of Inflection

Second Derivative Test for Local Extrema

Graphical Behavior of Functions from Derivatives

4.5 Indeterminate Forms and L'Hôpital's Rule

Indeterminate Form $0/0$

Indeterminate Forms ∞/∞ , $\infty \cdot 0$, $\infty - \infty$

Indeterminate Powers

Proof of L'Hôpital's Rule

4.6 Applied Optimization

Examples from Mathematics and Physics

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4.8 Antiderivatives

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5.1 Area and Estimating with Finite Sums

Area

Distance Traveled

Displacement Versus Distance Traveled

Average Value of a Nonnegative Continuous Function

5.2 Sigma Notation and Limits of Finite Sums

Finite Sums and Sigma Notation

Limits of Finite Sums

Riemann Sums

5.3 The Definite Integral

Definition of the Definite Integral

Integrable and Nonintegrable Functions

The Midpoint Rule

Properties of Definite Integrals

Area Under the Graph of a Nonnegative Function

Average Value of a Continuous Function Revisited

5.4 The Fundamental Theorem of Calculus

Mean Value Theorem for Definite Integrals

Fundamental Theorem, Part 1

Fundamental Theorem, Part 2 (The Evaluation Theorem)

The Integral of a Rate

The Relationship Between Integration and Differentiation

Total Area

5.5 Indefinite Integrals and the Substitution Method

Substitution: Running the Chain Rule Backwards

Trying Different Substitutions

5.6 Definite Integral Substitutions and the Area Between Curves

The Substitution Formula

Definite Integrals of Symmetric Functions

Areas Between Curves

Integration with Respect to y

Applications of Definite Integrals

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6.1 Volumes Using Cross-Sections

NOTE Disk and washer methods, plus solids with known cross-sections. The AP exam does not require cylindrical shells (6.2).

Slicing by Parallel Planes

Solids of Revolution: The Disk Method

Solids of Revolution: The Washer Method

6.3 Arc Length

Length of a Curve $y = f(x)$

Dealing with Discontinuities in dy/dx

The Differential Formula for Arc Length

Integrals and Transcendental Functions

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7.2 Exponential Change and Separable Differential Equations

NOTE Separable differential equations and exponential growth and decay. Logistic growth is in 16.4.

Exponential Change

Separable Differential Equations

Unlimited Population Growth

Radioactivity

Heat Transfer: Newton's Law of Cooling

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8.1 Using Basic Integration Formulas

8.2 Integration by Parts

Product Rule in Integral Form

Evaluating Definite Integrals by Parts

8.5 Integration of Rational Functions by Partial Fractions

NOTE AP only tests partial fractions with distinct linear factors in the denominator.

General Description of the Method

Determining Coefficients by Differentiating

8.7 Numerical Integration

NOTE The Trapezoidal Rule is tested; Simpson's Rule is not.

Approximating Integrals with the Midpoint Rule

Trapezoidal Approximations

Simpson's Rule: Approximations Using Parabolas

Error Analysis

8.8 Improper Integrals

Infinite Limits of Integration

The Integral $\int_1^{\infty} dx/x^p$

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Improper Integrals with a CAS

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9.1 Sequences

Representing Sequences

Convergence and Divergence

Calculating Limits of Sequences

Using L'Hôpital's Rule

Commonly Occurring Limits

Recursive Definitions

Bounded Monotonic Sequences

9.2 Infinite Series

Geometric Series

The n th-Term Test for a Divergent Series

Combining Series

Adding or Deleting Terms

Reindexing

9.3 The Integral Test

The Integral Test

Nondecreasing Partial Sums

Error Estimation

9.4 Comparison Tests

The Limit Comparison Test

9.5 Absolute Convergence; The Ratio and Root Tests

NOTE The Ratio Test is tested; the Root Test is not.

The Ratio Test

The Root Test

9.6 Alternating Series and Conditional Convergence

Conditional Convergence

Rearranging Series

Summary of Tests to Determine Convergence or Divergence

9.7 Power Series

Power Series and Convergence

The Radius of Convergence of a Power Series

Operations on Power Series

9.8 Taylor and Maclaurin Series

Series Representations

Taylor and Maclaurin Series

Taylor Polynomials

9.9 Convergence of Taylor Series

NOTE Emphasize the Lagrange Error Bound (Taylor's Theorem with remainder).

Estimating the Remainder

Using Taylor Series

A Proof of Taylor's Theorem

9.10 Applications of Taylor Series

The Binomial Series for Powers and Roots

Evaluating Nonelementary Integrals

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10.1 Parametrizations of Plane Curves

Parametric Equations

Cycloids

Brachistochrones and Tautochrones

10.2 Calculus with Parametric Curves

Tangent Lines and Areas

Length of a Parametrically Defined Curve

Length of a Curve $y = f(x)$

The Arc Length Differential

Areas of Surfaces of Revolution

10.3 Polar Coordinates

Definition of Polar Coordinates

Polar Equations and Graphs

Relating Polar and Cartesian Coordinates

10.4 Graphing Polar Coordinate Equations

Symmetry

Slope

Converting a Graph from the $r\theta$ -Plane to the xy -Plane

10.5 Areas and Lengths in Polar Coordinates

NOTE AP tests area in polar coordinates. Polar arc length is not tested.

Area in the Plane

Length of a Polar Curve

CHAPTER 12

Vector-Valued Functions and Motion in Space

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12.1 Curves in Space and Their Tangents

NOTE AP covers plane curves only. Treat vector-valued functions as $\langle x(t), y(t) \rangle$.

Limits and Continuity

Derivatives and Motion

Differentiation Rules

Vector Functions of Constant Length

12.2 Integrals of Vector Functions; Projectile Motion

NOTE Focus on recovering position from velocity and on displacement vs. total distance traveled.

Integrals of Vector Functions

The Vector and Parametric Equations for Ideal Projectile Motion

Projectile Motion with Wind Gusts

First-Order Differential Equations

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16.1 Solutions, Slope Fields, and Euler's Method

NOTE Slope fields and Euler's method.

General First-Order Differential Equations and Solutions

Slope Fields: Viewing Solution Curves

Euler's Method

16.4 Graphical Solutions of Autonomous Equations

NOTE Logistic growth: the carrying capacity and where the population grows fastest.

Equilibrium Values and Phase Lines

Stable and Unstable Equilibria

Newton's Law of Cooling

A Falling Body Encountering Resistance

The Logistic Model for Population Growth

The Logistic Equation in Neural Networks and Machine Learning